

TADEBYAYAKHA, Russian Federation

WMO: 209640

Lat: 70.350N

Lon: 74.133E

Elev: 13

StdP: 14.69

Time Zone: 5.00 (E05)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-45.3	-40.7	-48.0	0.3	-44.2	-43.9	0.4	-40.1	36.9	-0.9	32.8	-0.3	5.8	70	1.079

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	67.2	58.9	62.0	56.4	57.2	53.3	59.7	65.8	56.6	61.2	53.4	57.1	13.3	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.6	68.3	62.3	53.9	61.7	59.4	50.9	55.2	56.4	26.3	66.4	24.2	61.2	22.3	57.1	70.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.6	29.1	26.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	-16.6	-12.9	-4.3	0.3	19.9	35.5	46.8	45.7	38.0	18.9	0.4	-9.6
	DBStd	25.86	15.27	16.54	16.14	15.04	8.94	7.47	8.62	5.15	5.73	12.72	16.12	15.78
	HDD50	13366	2063	1761	1683	1492	934	443	176	154	362	963	1487	1848
	HDD65	18736	2528	2181	2148	1942	1399	884	567	598	811	1428	1937	2313
	CDD50	107	0	0	0	0	0	10	75	21	1	0	0	0
	CDD65	2	0	0	0	0	0	0	2	0	0	0	0	0
	CDH74	21	0	0	0	0	0	1	19	1	0	0	0	0
	CDH80	3	0	0	0	0	0	0	3	0	0	0	0	0
	WSAvg	12.6	13.2	12.9	13.2	12.5	13.1	10.7	10.0	11.7	12.4	13.9	13.3	13.8
(15) Precipitation	PrecAvg	23.7	2.2	2.0	1.7	1.7	1.6	1.7	1.6	2.0	2.2	2.6	2.2	2.3
	PrecMax	30.6	5.3	5.3	4.5	3.1	2.8	3.1	3.3	3.9	3.9	3.8	5.4	4.0
	PrecMin	17.5	0.4	0.4	0.4	0.3	0.9	0.3	0.4	0.6	0.7	1.4	0.4	1.0
	PrecStd	3.6	1.0	1.0	0.8	0.7	0.5	0.7	0.8	0.7	0.8	0.6	0.8	0.8
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.2	27.1	31.6	34.0	37.6	67.5	76.0	66.8	55.4	42.1	33.0	28.8
		MCWB	20.8	26.6	31.3	32.8	36.4	57.1	62.9	60.3	49.7	40.5	32.1	27.9
	2%	DB	16.3	22.1	28.4	32.5	34.9	57.6	70.1	60.2	50.8	38.8	31.2	24.0
		MCWB	15.7	21.5	28.0	31.7	33.6	52.6	60.7	55.4	48.5	37.4	30.7	23.5
	5%	DB	10.9	17.8	24.9	30.5	33.7	50.1	65.9	56.6	48.3	36.6	28.8	18.9
		MCWB	10.2	17.1	24.5	29.7	32.6	46.9	58.1	53.1	46.6	35.5	28.1	18.2
	10%	DB	6.1	11.5	20.1	25.9	32.6	44.8	61.6	53.9	46.0	34.5	26.0	14.8
		MCWB	5.4	10.9	19.4	25.0	31.6	42.2	56.2	51.0	44.3	33.4	25.4	14.2
	0.4%	WB	21.0	26.7	31.3	33.1	36.7	58.0	65.2	60.2	51.0	41.0	32.3	28.1
		MCDB	21.1	26.9	31.4	33.7	37.5	65.4	72.3	65.5	53.5	42.1	32.5	28.7
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	15.8	21.8	28.0	31.9	33.9	52.6	61.9	56.4	48.8	37.7	30.6	23.4
		MCDB	16.0	22.2	28.2	32.4	34.4	57.9	69.4	59.1	50.5	38.7	31.0	23.8
	5%	WB	10.5	17.2	24.5	29.9	32.8	46.5	59.0	53.8	46.7	35.7	28.2	18.4
		MCDB	11.1	17.7	24.9	30.7	33.6	50.0	64.7	56.1	48.1	36.4	28.6	19.0
	10%	WB	5.4	11.0	19.5	25.1	31.7	42.3	56.3	51.2	44.5	33.5	25.4	14.2
		MCDB	6.0	11.5	20.0	26.0	32.6	44.7	61.2	53.4	45.9	34.3	26.0	14.8
	5% DB	MDBR	14.1	15.0	16.1	16.8	10.0	8.2	12.0	8.7	7.4	10.2	13.4	14.2
		MCDBR	22.0	18.2	17.8	13.9	7.1	17.8	20.2	15.0	10.1	7.0	14.3	18.1
		MCWBR	21.9	17.9	17.8	13.5	7.0	12.6	11.7	10.6	8.4	6.9	14.2	18.0
		MCWBR	21.9	18.4	17.0	13.1	7.6	17.1	18.9	13.2	9.3	7.2	14.2	18.1
(40) Clear-Sky Solar Irradiance	taub		N/A	0.237	0.294	0.316	0.336	0.349	0.375	0.358	0.312	0.235	N/A	N/A
		taud	N/A	1.955	2.027	2.061	2.100	2.310	2.341	2.383	2.415	2.104	N/A	N/A
	Ebn at Noon		0	140	214	253	264	266	253	242	221	143	0	0
		Edh at Noon	0	13	27	36	39	33	31	27	19	11	0	0
(44) All-Sky Solar Radiation	RadAvg	0	79	545	1237	1835	1811	1571	979	453	133	0	0	0
	RadStd	0	11	51	113	92	129	199	124	47	19	0	0	0

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page